MISSIONARIES AND CANNIBALS PROBLEM

Aim

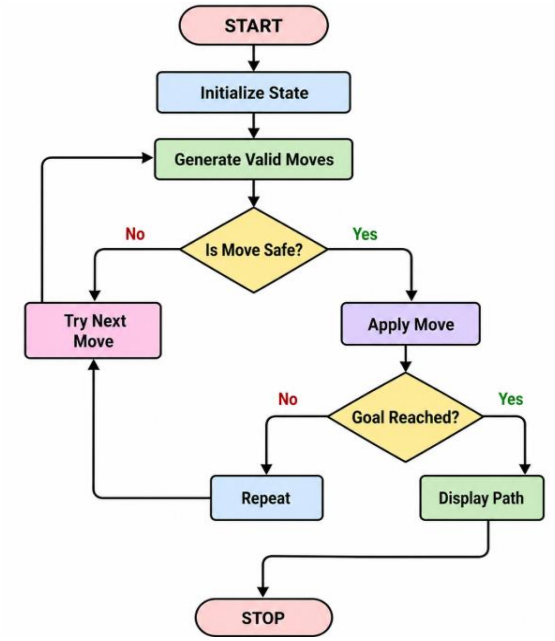
To solve the Missionaries and Cannibals problem using Artificial Intelligence search techniques.

Objectives

- To understand state-space search.
- To transport all missionaries and cannibals safely.
- To ensure missionaries are never outnumbered.
- To find a valid sequence of moves.

Algorithm

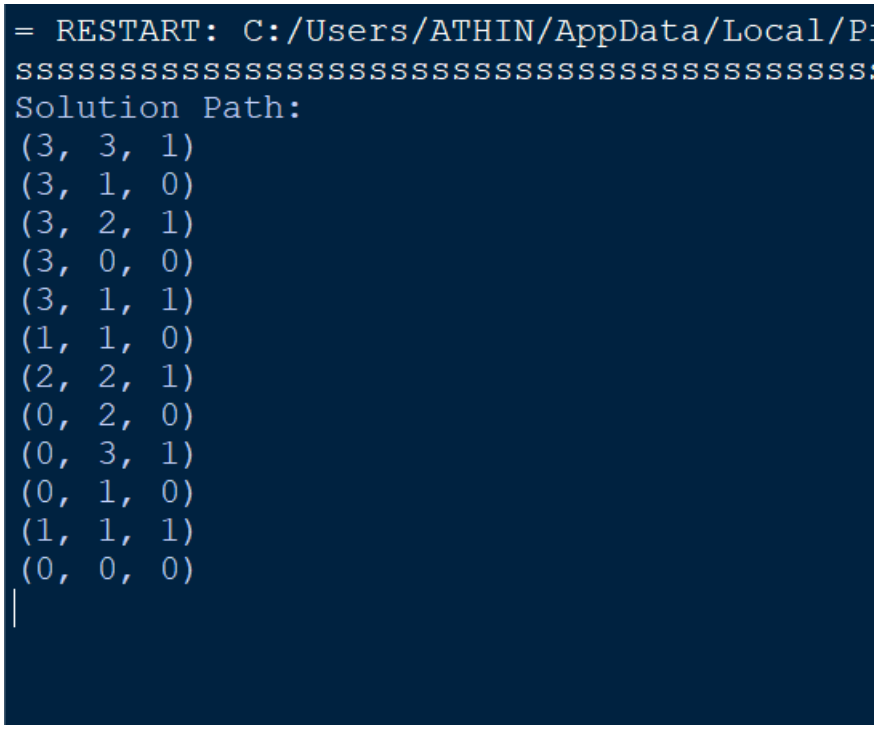
1. Start.
2. Initialize the starting state (3,3,1).
3. Generate all possible valid moves.
4. Check whether the move is safe.
5. If safe, move missionaries and/or cannibals.
6. Repeat until the goal state (0,0,0) is reached.
7. Display the solution path.
8. Stop.



PROGRAM

```
from collections import deque

def missionaries_cannibals():
    start = (3, 3, 1)
    goal = (0, 0, 0)

    queue = deque([(start, [])])
    visited = set()

    moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        if state in visited:
            continue

        visited.add(state)

        m, c, boat = state

        for dm, dc in moves:
            if boat:
                new = (m-dm, c-dc, 0)
            else:
                new = (m+dm, c+dc, 1)

            nm, nc, nb = new

            if (0 <= nm <= 3 and 0 <= nc <= 3 and
                (nm == 0 or nm >= nc) and
                (3-nm == 0 or 3-nm >= 3-nc)):

                queue.append((new, path + [state]))

solution = missionaries_cannibals()

print("Solution Path:")
```

OUTPUT



Result

The Missionaries and Cannibals problem was successfully solved using the Breadth First Search (BFS) algorithm, and all missionaries and cannibals were transported safely to the opposite bank.